



Belgavi
Mini Project Synopsis

“RADAR SURVEILLANCE SYSTEM ”

submitted to Visvesvaraya Technological University in partial fulfillment for the completion of Mini project in 5th semester B.E. , Electronics and Communication Engineering.

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September 2024

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1.ABSTRACT

RADAR is a detection system that uses radio waves to determine the characteristics of the detected object .In this project the ultrasonic sensor is used to measure the distance between the radar and the object and also speed of the object. The movement of the sensor is controlled by servo motor. The signal received from the sensor would be processed, then the result would be shown on a serial monitor. RADAR surveillance system is suitable for various applications Including border security, military surveillance , air traffic control and industrial automation. The system's open architecture allows seamless integration with existing security infrastructure and facilitates data analysis for intelligent decision-making.

2.INTRODUCTION

RADAR stands for radio Detection and ranging system. RADAR is an object detection system which uses radio waves to determine objects location, distance. It is basically an electromagnetic system used to detect the location and distance of an object from point where radar is used. It works by radiating waves into space and monitoring the echo from the object. Some radar system are also used in air traffic control at airport. Its primary purpose is to enhance security, surveillance, and situational awareness, with applications in border security, military surveillance, air traffic control, industrial automation, traffic management, and weather monitoring. Key features include long-range detection , high accuracy and resolution, all-weather capability, and integration with existing security systems. The RSS consists of radar antenna, transmitter and receiver, signal processing unit, display and control unit, and communication interface. Technological advancements include advanced signal processing algorithms, high-resolution radar imaging.

3. Literature Survey

The literature survey on RADAR surveillance system highlights the several key features and benefits. Here are some main points.

Long-Range Detection Radar Surveillance Systems can detect targets at ranges of up to several kilometers, making them effective for border security, military surveillance, and air traffic control .High Accuracy and Resolution Advanced signal processing algorithms and phased array radar technology enable high accuracy and resolution, allowing for precise target tracking and classification [1].

All-Weather Capability Radar Surveillance Systems can operate effectively in adverse weather conditions, including rain, fog, and snow, ensuring continuous surveillance Integration with AI and Machine Learning Modern Radar Surveillance Systems integrate with AI and machine learning algorithms for enhanced target detection, classification, and tracking [2].

Enhanced Security and Surveillance Radar Surveillance Systems provide real-time information on target location, speed, and direction, enhancing security and situational awareness.Improved Efficiency and Productivity Automated target detection and tracking reduce manual errors and increase operator efficiency [3].

Cost-Effective Radar Surveillance Systems offer a cost-effective solution for long-range surveillance, reducing the need for multiple sensors and infrastructure.Real-Time Data for Decision-Making Radar Surveillance Systems provide real-time data for intelligent decision-making, enabling quick response to security threats [4]

4.Aim ,Objectives and methodology

Aim:

Detect and track the moving objects within specified range and provide information Of object location, speed and distance

Objectives:

The main objective of RADAR surveillance system is to accurately detect and locate objects .Additionally system can be used for security, surveillance and automation purpose.

Methodology:

1. Ultrasonic sensors generates high frequency sound waves and evaluate the echo which is received back by the sensor.
2. Sensors calculate the time interval between sending the signal and receiving the echo to determine the distance of an object.
3. System typically use a transducer which generates sound waves in the ultrasonic range, above 18,000hertz,by converting electrical energy into sound.
4. Then upon receiving the echo convert the sound waves into electrical energy which can be measured and displayed.

5. BLOCK DIAGRAM



Figure: Block diagram of RADAR system

6.WORKING PRINCIPLE

1. The working principle of radar involves transmitting electromagnetic waves through an antenna, which reflect off objects and return to the radar system.
2. The receiver processes the returned signals, measuring time delay and frequency shift to calculate object distance, speed, and direction.
3. Radar uses the Doppler shift principle to detect motion and the radar equation to determine signal strength and detection range.
4. There are various types of radar, including pulse radar, continuous wave radar, and phased array radar. operating on different frequency bands such as X-band, S-band, and L-band.

7. HARDWARE /SOFTWARE REQUIREMENTS:

HARDWARE:

1. Arduino uno
2. Ultrasonic sensor
3. Servo motor
4. Buzzer
5. Bluetooth cable
6. LED'S
7. Resistors
8. Breadboard

Software:

- Arduino IDE

8.Expected outcomes:

1. Output will be displayed on the serial monitor of PC.
2. Experimental result shows that system can detect object's distance, angle, speed and direction (towards or away) within the range

9.REFERENCES:

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